

Supplemental material for Designing Frustrated Quantum Magnets with Laser-Dressed Rydberg Atoms

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(Dated: February 2, 2015)

I. VAN DER WAALS INTERACTIONS BETWEEN $j = 1/2$ RYDBERG STATES

Away from Foerster resonances two laser excited Rydberg atoms dominantly interact via van der Waals interactions [1]. These van der Waals interactions, $H_{\text{vdW}}^{(i,k)}$, will mix different Zeeman sublevels $|m_j\rangle$ in the $nP_{1/2}$ manifold [2]. Let us denote by $\hat{P} = \sum_{i,j} |m_i, m_j\rangle \langle m_i, m_j|$ a projection operator into the $nP_{1/2}$ manifold, then dipole-dipole interactions

$$V_{\text{dd}}^{(i,k)}(\mathbf{r}) = -\sqrt{\frac{24\pi}{5}} \frac{1}{r^3} \sum_{\mu,\nu} C_{\mu,\nu;\mu+\nu}^{1,1;2} Y_2^{\mu+\nu}(\vartheta, \varphi)^* d_\mu^{(i)} d_\nu^{(j)},$$

will couple states in the $\hat{P}$ manifold to intermediate states, $\hat{Q}_{\alpha,\beta} = |\alpha, \beta\rangle \langle \alpha, \beta|$, which have an energy difference $\delta_{\alpha\beta}$. Here, $\mathbf{d}^{(i)}$ is the dipole operator of the i -th atom and $\mathbf{r} = (r, \vartheta, \varphi)$ is the relative vector between atom i and atom j in spherical coordinates and $d_\mu^{(i)}$ is the μ -th spherical components ($\mu, \nu \in \{-1, 0, 1\}$) of the atomic dipole operator. With $C_{m_1, m_2; M}^{j_1, j_2; J}$ we denote the Clebsch-Gordan coefficients and Y_l^m are spherical harmonics. In second order perturbation theory this gives rise to

$$H_{\text{vdW}}^{(i,k)} = \hat{P} \sum_{\alpha\beta} \frac{V_{\text{dd}}^{(i,k)} \hat{Q}_{\alpha,\beta} V_{\text{dd}}^{(i,k)}}{\delta_{\alpha\beta}} \hat{P}, \quad (1)$$

where $H_{\text{vdW}}^{(i,k)}$ is understood as an operator acting in the manifold of Zeeman sublevels.

Due to the odd parity of the electric dipole operators $d_\mu^{(i)}$ and $d_\nu^{(j)}$, the dipole-dipole interaction, V_{dd} , can couple initial $nP_{1/2}$ states only to $n'S_{1/2}$ or $n''D_{3/2}$

states. Therefore, there are four possible channels shown in Tab. I(left) for which the matrix element $\langle nP_{1/2} m_1 | \langle nP_{1/2} m_1 | V_{\text{dd}} | n', \ell_\alpha, j_\alpha, m_\alpha \rangle | n'', \ell_\beta, j_\beta, m_\beta \rangle$ of Eq. (1) is non-zero. Here, $(\ell_{\alpha,\beta}, j_{\alpha,\beta})$ can either correspond to $S_{1/2}$ or $D_{3/2}$ states depending on the channel. While there is no selection rule for possible final principal quantum numbers n' and n'' which solely determine the overall strength of the matrix element, the dipole-dipole matrix element is only non-zero if the magnetic quantum numbers and the spherical component of the dipole operator fulfill $m_1 + \mu = m_\alpha$ and $m_2 + \nu = m_\beta$. The total vdW interaction of Eq. (1) can be obtained by summing over all channels ν , that is

$$\hat{V}_{\text{vdW}} = \sum_{\nu} C_6^{(\nu)} \mathcal{D}_\nu(\vartheta, \varphi) / r^6. \quad (2)$$

Here, $C_6^{(\nu)}$ contains the radial part of the matrix elements

$$C_6^{(\nu)} = \sum_{n_\alpha, n_\beta} \frac{\mathcal{R}_1^\alpha \mathcal{R}_2^\beta \mathcal{R}_3^\alpha \mathcal{R}_4^\beta}{\delta_{\alpha\beta}} \quad (3)$$

which accounts for the overall strength of the interaction and is independent of the magnetic quantum numbers. With $\mathcal{R}_i^k = \int dr r^2 \psi_{n_i, \ell_i, j_i}(r)^* r \psi_{n_k, \ell_k, j_k}(r)$ we denote the radial integral. The matrix

$$\mathcal{D}_\nu(\vartheta, \varphi) = \hat{P}_{12} \sum_{m_\alpha, m_\beta} \mathcal{M}_\nu \hat{Q}_{\alpha,\beta} \mathcal{M}_\nu \hat{P}_{34} \quad (4)$$

on the other hand is a matrix in the subspace of magnetic quantum numbers which contains the relative angles between the two atoms ($s = 1/2$)

$$\begin{aligned} \langle m_1, m_2 | \mathcal{M}_\nu | m_\alpha, m_\beta \rangle = & (-)^{s-m_1} \sqrt{(2\ell_1+1)(2j_1+1)(2\ell_\alpha+1)(2j_\alpha+1)} \begin{Bmatrix} \ell_1 & \ell_\alpha & 1 \\ j_\alpha & j_1 & s \end{Bmatrix} \begin{pmatrix} \ell_\alpha & 1 & \ell_1 \\ 0 & 0 & 0 \end{pmatrix} \\ & \times (-)^{s-m_2} \sqrt{(2\ell_2+1)(2j_2+1)(2\ell_\beta+1)(2j_\beta+1)} \begin{Bmatrix} \ell_2 & \ell_\beta & 1 \\ j_\beta & j_2 & s \end{Bmatrix} \begin{pmatrix} \ell_\beta & 1 & \ell_2 \\ 0 & 0 & 0 \end{pmatrix} \\ & \times \left[-\sqrt{\frac{24\pi}{5}} \sum_{\mu,\nu} C_{\mu,\nu;\mu+\nu}^{1,1;2} \begin{pmatrix} j_\alpha & 1 & j_1 \\ m_\alpha & \mu & -m_1 \end{pmatrix} \begin{pmatrix} j_\beta & 1 & j_2 \\ m_\beta & \nu & -m_2 \end{pmatrix} Y_2^{\mu+\nu}(\vartheta, \varphi)^* \right]. \end{aligned} \quad (5)$$

For the individual channels $\nu \in \{a, b, c, d\}$ of Tab. I we find

$$\begin{aligned} \text{(a)} \quad \mathcal{D}_a(\vartheta, \varphi) &= \frac{2}{9} \mathbb{1}_4 - \mathcal{D}_0(\vartheta, \varphi), \\ \text{(b)} \quad \mathcal{D}_b(\vartheta, \varphi) &= \frac{4}{9} \mathbb{1}_4 - \mathcal{D}_0(\vartheta, \varphi), \\ \text{(c, d)} \quad \mathcal{D}_c(\vartheta, \varphi) &= \mathcal{D}_d(\vartheta, \varphi) = \mathcal{D}_0(\vartheta, \varphi), \end{aligned} \quad (6)$$

with $\mathbb{1}_4$ the 4×4 identity matrix and

$$\mathcal{D}_0(\vartheta, \varphi) = \frac{1}{81} \begin{pmatrix} 3 \cos(2\vartheta) + 11 & 3e^{-i\varphi} \sin(2\vartheta) & 3e^{-i\varphi} \sin(2\vartheta) & 6e^{-2i\varphi} \sin^2(\vartheta) \\ 3e^{i\varphi} \sin(2\vartheta) & 13 - 3 \cos(2\vartheta) & -3 \cos(2\vartheta) - 5 & -3e^{-i\varphi} \sin(\vartheta) \\ 3e^{i\varphi} \sin(2\vartheta) & -3 \cos(2\vartheta) - 5 & 13 - 3 \cos(2\vartheta) & -3e^{-i\varphi} \sin(\vartheta) \\ 6e^{2i\varphi} \sin^2(\vartheta) & -3e^{i\varphi} \sin(2\vartheta) & -3e^{i\varphi} \sin(2\vartheta) & 3 \cos(2\vartheta) + 11 \end{pmatrix} \quad (7)$$

written in the basis $\{|\frac{1}{2}, -\frac{1}{2}\rangle, |-\frac{1}{2}, \frac{1}{2}\rangle, |\frac{1}{2}, \frac{1}{2}\rangle, |-\frac{1}{2}, -\frac{1}{2}\rangle\}$ of Zeeman states in the $j = 1/2$ Rydberg manifold. For the special orientations (i) $\vartheta = 0$ we find

$$\mathcal{D}_0(0, 0) = \frac{1}{81} \begin{pmatrix} 14 & 0 & 0 & 0 \\ 0 & 10 & -8 & 0 \\ 0 & -8 & 10 & 0 \\ 0 & 0 & 0 & 14 \end{pmatrix} \quad (8)$$

and (ii) for $\vartheta = \pi/2$ the matrix simplifies to

$$\mathcal{D}_0(\frac{\pi}{2}, 0) = \frac{1}{81} \begin{pmatrix} 8 & 0 & 0 & 6 \\ 0 & 16 & -2 & 0 \\ 0 & -2 & 16 & 0 \\ 6 & 0 & 0 & 8 \end{pmatrix}. \quad (9)$$

The total vdW interaction matrix in the $nP_{1/2}$ subspace becomes

$$H_{\text{vdW}}^{(i,k)} = \frac{2}{9} \left(C_6^{(a)} + 2C_6^{(b)} \right) \mathbb{1}_4 + \left(2C_6^{(c)} - C_6^{(a)} - C_6^{(b)} \right) \mathcal{D}_0, \quad (10)$$

where the coefficients $C_6^{(\nu)}$ depend on the principal quantum number n , see Fig. 1.

We note that the vdW Hamiltonian describing the interactions between $S_{1/2}$ -states can be written in the exact same form as Eq. (10). However, the coupling terms, $C_6^{(\nu)}$, correspond to the channels of Tab. I(right). Therefore, the radial matrix elements for $S_{1/2}$ states differ only slightly due to the fine structure splitting ΔE_{FS} between $P_{1/2}$ and $P_{3/2}$ states, see Fig. 1(right). In the limit where the fine structure can be neglected compared to other energy scales we find $C_6^{(a)} = C_6^{(b)} = C_6^{(c)} = C_6^{(d)}$ and the vdW interaction of Eq. (10) between $nS_{1/2}$ states becomes diagonal, that is $H_{\text{vdW}}^{(i,k)} = (2/3)C_6^{(a)}\mathbb{1}_4$. Thus, there is no vdW mixing between Zeeman sublevels if the fine-structure splitting can be neglected. This can be understood by a simple argument: Since for s -states the different m_j levels are proportional to the electronic spin

m_s , that is $|m_j = \pm \frac{1}{2}\rangle = |\ell = 0, m_\ell = 0\rangle \otimes |m_s = \pm \frac{1}{2}\rangle$, and since dipole-dipole interactions cannot mix spin degrees of freedom there cannot be any vdW mixing of Zeeman levels in the absence of fine-structure. The first correction will be proportional to $\sim \Delta E_{FS}/\delta_{\alpha\beta}\mathcal{D}_0$. It is therefore only the spin-orbit coupling in the intermediate $Q_{\alpha,\beta}$ manifold which mixes different Zeeman sublevels in the case of $S_{1/2}$ states.

On the contrary, for $P_{1/2}$ states, the radial coefficients $C_6^{(\nu)}$ differ much more strongly due to the energy difference between d - and s -states and due to the fact that Zeeman sublevels in the $nP_{1/2}$ manifold are already a superposition between $m_s = \pm \frac{1}{2}$ states of the electronic spin. Therefore, mixing of Zeeman sublevels for $nP_{1/2}$ states can be of the same order of magnitude than the diagonal terms and play a significant role. In the special (1D) case $\vartheta = 0$, the doubly excited levels $|\frac{1}{2}, -\frac{1}{2}\rangle$ and $|\frac{1}{2}, \frac{1}{2}\rangle$ are not coupled to any other doubly excited states which is a consequence of the conservation of the total angular momentum. On the contrary, for $\vartheta = \pi/2$ (atoms polarized perpendicular to the plane), the Hamiltonian of Eq. (10) reduces to Eq. [2] in the main text, with

$$\begin{aligned} c_{++} &= \frac{2}{81} \left(5C_6^{(a)} + 14C_6^{(b)} + 8C_6^{(c)} \right), \\ c_{+-} &= \frac{2}{81} \left(C_6^{(a)} + 10C_6^{(b)} + 16C_6^{(c)} \right), \\ w_{++} &= -\frac{2}{27} \left(C_6^{(a)} + C_6^{(b)} - 2C_6^{(c)} \right), \\ w_{+-} &= \frac{2}{81} \left(C_6^{(a)} + C_6^{(b)} - 2C_6^{(c)} \right) = -\frac{w_{++}}{3}. \end{aligned} \quad (11)$$

shown in Fig. [2](a) of the main text as a function of the principal quantum number n . In the following sections of this supplemental material, we will consider this particular orientation as it is the simplest configuration of vdW coupling where the doubly laser-excited state $|\frac{1}{2}, -\frac{1}{2}\rangle$ is only coupled to $|\frac{1}{2}, \frac{1}{2}\rangle$.

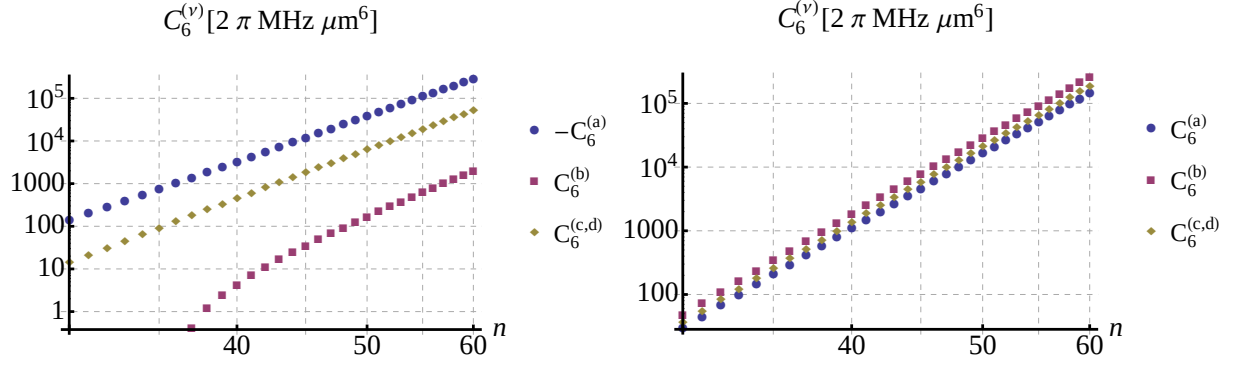


Figure 1. We plot the $C_6^{(\nu)}$ for (left) $nP_{1/2}$ and (right) $nS_{1/2}$ Rydberg states of ^{87}Rb as a function of the principal quantum number n for different channels ν of Tab. I.

ν	$(\ell, j) + (\ell, j) \longrightarrow (\ell_\alpha, j_\alpha) + (\ell_\beta, j_\beta)$
(a)	$P_{1/2} + P_{1/2} \longrightarrow S_{1/2} + S_{1/2}$
(b)	$P_{1/2} + P_{1/2} \longrightarrow D_{3/2} + D_{3/2}$
(c)	$P_{1/2} + P_{1/2} \longrightarrow S_{1/2} + D_{3/2}$
(d)	$P_{1/2} + P_{1/2} \longrightarrow D_{3/2} + S_{1/2}$

ν	$(\ell, j) + (\ell, j) \longrightarrow (\ell_\alpha, j_\alpha) + (\ell_\beta, j_\beta)$
(a)	$S_{1/2} + S_{1/2} \longrightarrow P_{1/2} + P_{1/2}$
(b)	$S_{1/2} + S_{1/2} \longrightarrow P_{3/2} + P_{3/2}$
(c)	$S_{1/2} + S_{1/2} \longrightarrow P_{1/2} + P_{3/2}$
(d)	$S_{1/2} + S_{1/2} \longrightarrow P_{3/2} + P_{1/2}$

Table I. Dipole-dipole interactions can couple $P_{1/2}$ (left) and $S_{1/2}$ (right) states to four channels (a-d).

II. LASER EXCITATION AND HYPERFINE GROUND STATES

The laser Hamiltonian, $H_L^{(i)}$, couples two hyperfine ground states $|g_- \rangle$ and $|g_+ \rangle$ to the Zeeman sublevels in

the $nP_{1/2}$ Rydberg manifold with detunings Δ_σ and Rabi frequencies Ω_σ ($\sigma = +, -$), respectively, see Fig. [1](b) in the main text. Uncoupling the nuclear spin the hyperfine ground states read

$$\begin{aligned}
 |g_+ \rangle &\equiv |5^2S_{1/2}, F=2, m_F=2 \rangle = |m_j = \tfrac{1}{2} \rangle |m_I = \tfrac{3}{2} \rangle, \\
 |g_- \rangle &\equiv |5^2S_{1/2}, F=1, m_F=1 \rangle = \frac{1}{2} \left[|m_j = \tfrac{1}{2} \rangle |m_I = \tfrac{1}{2} \rangle - \sqrt{3} |m_j = -\tfrac{1}{2} \rangle |m_I = \tfrac{3}{2} \rangle \right],
 \end{aligned}$$

where m_I is the projection quantum number of the nuclear spin. Using σ_+ and σ_- polarized light for the transition

$$\begin{aligned}
 |g_- \rangle &\xrightarrow{\Omega_+, \sigma_+} |nP_{1/2}, m_j = +\tfrac{1}{2} \rangle \otimes |m_I = \tfrac{3}{2} \rangle, \\
 |g_+ \rangle &\xrightarrow{\Omega_-, \sigma_-} |nP_{1/2}, m_j = -\tfrac{1}{2} \rangle \otimes |m_I = \tfrac{3}{2} \rangle,
 \end{aligned} \tag{12}$$

respectively, couples to two different Rydberg states but both in the same nuclear state. Thus, hyperfine structure can be treated as a spectator in the Rydberg manifold. Neglecting (small) hyperfine interactions, these are closed cycle transitions and do not couple to any other states in the hyperfine manifold. There are several alternative possibilities, e.g.

$$\begin{aligned}
 |g_+ \rangle &\equiv |5^2S_{1/2}, F=2, m_F=1 \rangle = \frac{1}{2} \left[\sqrt{3} |m_j = \tfrac{1}{2} \rangle |m_I = \tfrac{1}{2} \rangle + |m_j = -\tfrac{1}{2} \rangle |m_I = \tfrac{3}{2} \rangle \right], \\
 |g_- \rangle &\equiv |5^2S_{1/2}, F=1, m_F=0 \rangle = \frac{1}{\sqrt{2}} \left[|m_j = \tfrac{1}{2} \rangle |m_I = -\tfrac{1}{2} \rangle - |m_j = -\tfrac{1}{2} \rangle |m_I = \tfrac{1}{2} \rangle \right],
 \end{aligned}$$

which can be laser excited to specific Rydberg states

$$\begin{aligned}
 |g_- \rangle &\xrightarrow{\Omega_+, \sigma_+} |nP_{1/2}, m_j = -\tfrac{1}{2} \rangle \otimes |m_I = \tfrac{1}{2} \rangle, \\
 |g_+ \rangle &\xrightarrow{\Omega_-, \sigma_-} |nP_{1/2}, m_j = +\tfrac{1}{2} \rangle \otimes |m_I = \tfrac{1}{2} \rangle.
 \end{aligned}$$

III. EFFECTIVE GROUND STATE POTENTIALS

A. Adiabatic elimination

In the dressing limit, $\Omega_\sigma \ll \Delta_{\sigma'}$, atoms initially in their electronic ground states $|g\rangle_1 \dots |g\rangle_N$ are off-resonantly

coupled to the Rydberg states $|r\rangle_1 \dots |r\rangle_N$ and the new “dressed” ground states inherit a tunable fraction of the Rydberg interaction [3]. The effective interaction potential between N atoms in their dressed ground states, $|\tilde{g}\rangle_1 \dots |\tilde{g}\rangle_N$, can be obtained by diagonalizing the Hamiltonian H_{mic} for a fixed relative position and zero kinetic energy. The total Hamiltonian H_{mic} has block structure

$$H_{\text{mic}} = \begin{pmatrix} \mathbf{H}_0 & \mathbf{\Omega}_1 & 0 & 0 \\ \mathbf{\Omega}_1^\dagger & \mathbf{H}_1 & \mathbf{\Omega}_2 & 0 \\ 0 & \mathbf{\Omega}_2^\dagger & \mathbf{H}_2 & \mathbf{\Omega}_3 \\ 0 & 0 & \mathbf{\Omega}_3^\dagger & \mathbf{H}_3 \\ & & & \ddots \end{pmatrix} \quad (13)$$

where $\mathbf{H}_n$ governs the dynamics in the subspace with n -Rydberg excitations present, while the $\mathbf{\Omega}_n$ matrices describe the coupling between adjacent sectors n and $n-1$ due to the laser. Only subspaces $\mathbf{H}_{n \geq 2}$ contain the interaction potentials V_{ij} and W_{ij} since we assume that ground and Rydberg states do not interact at long distances.

As a first step we diagonalize $H_A^{(1)} + H_A^{(2)} + H_{\text{vdW}}$ as the dominant part of the Hamiltonian in the subspace of two Rydberg excitations. This results in four new eigenstates

$$\begin{aligned} |E_{\sigma\sigma}(\rho)\rangle &= [\cos \chi(\rho) |r_\sigma r_\sigma\rangle + \sigma \sin \chi(\rho) |r_{\bar{\sigma}} r_{\bar{\sigma}}\rangle] / \sqrt{2}, \\ |E_{\sigma\bar{\sigma}}(\rho)\rangle &= (|r_\sigma r_{\bar{\sigma}}\rangle + \sigma |r_{\bar{\sigma}} r_\sigma\rangle) / \sqrt{2} \end{aligned}$$

where $\sigma = +, -$ and we defined $\bar{\sigma} \equiv -\sigma$ with correspond-

ing eigenenergies

$$\begin{aligned} E_{\sigma\sigma}(\rho) &= V_{++}(\rho) - \Delta_{+-} + \sigma \sqrt{\delta_{+-}^2 + W_{++}(\rho)^2}, \\ E_{\sigma\bar{\sigma}}(\rho) &= V_{+-}(\rho) - \Delta_{+-} + \sigma W_{+-}(\rho), \end{aligned} \quad (14)$$

shown in Fig. [2](c) in the main text. Here, we used the short hand notation $\Delta_{+-} \equiv \Delta_+ + \Delta_-$, $\delta_{+-} \equiv \Delta_+ - \Delta_-$ and $\tan 2\chi(\rho) = W_{++}(\rho)/\delta_{+-}$.

Adiabatically eliminating (up to fourth order in $\Omega_\sigma/E_{\sigma'\sigma''} \ll 1$) of the Rydberg states yields an effective interaction in the subspace of hyperfine ground states

$$\begin{aligned} \tilde{H} &= \mathbf{H}_0 + \mathbf{H}_1 - \mathbf{\Omega}_1 \mathbf{H}_1^{-1} \mathbf{\Omega}_1^\dagger \\ &\quad + \mathbf{\Omega}_1 \mathbf{H}_1^{-1} \mathbf{\Omega}_1 \mathbf{H}_1^{-1} \mathbf{\Omega}_1^\dagger \mathbf{H}_1^{-1} \mathbf{\Omega}_1^\dagger \\ &\quad - \mathbf{\Omega}_1 \mathbf{H}_1^{-1} \mathbf{\Omega}_2 \mathbf{H}_2^{-1} \mathbf{\Omega}_2^\dagger \mathbf{H}_1^{-1} \mathbf{\Omega}_1^\dagger \end{aligned} \quad (15)$$

which yields (for two atoms)

$$\tilde{H} = \begin{pmatrix} \tilde{V}_{++} & 0 & 0 & \tilde{W}_{++} \\ 0 & \tilde{V}_{+-} & \tilde{W}_{+-} & 0 \\ 0 & \tilde{W}_{+-}^* & \tilde{V}_{+-} & 0 \\ \tilde{W}_{++}^* & 0 & 0 & \tilde{V}_{--} \end{pmatrix} \quad (16)$$

written in the basis of the hyperfine states $\{|g_+g_+\rangle, |g_-g_+\rangle, |g_+g_-\rangle, |g_-g_-\rangle\}$. In the following we will discuss the various potentials separately.

B. The potential $\tilde{V}_{++}$ and $\tilde{V}_{--}$

Adiabatic elimination up to fourth order in Ω/Δ of the Rydberg states yields

$$\begin{aligned} \tilde{V}_{++} &= \frac{\Omega_-^2}{2\Delta_-} - \frac{\Omega_-^4}{4\Delta_-^3} + \frac{\Omega_-^4}{4\Delta_-^2} \frac{V_{++} - 2\Delta_+}{W_{++}^2 - (V_{++} - 2\Delta_-)(V_{++} - 2\Delta_+)}, \\ \tilde{V}_{--} &= \frac{\Omega_+^2}{2\Delta_+} - \frac{\Omega_+^4}{4\Delta_+^3} + \frac{\Omega_+^4}{4\Delta_+^2} \frac{V_{++} - 2\Delta_-}{W_{++}^2 - (V_{++} - 2\Delta_-)(V_{++} - 2\Delta_+)}, \end{aligned}$$

where asymptotically we just recover the single particle light shifts (up to fourth order)

$$\begin{aligned} \tilde{V}_{++}^\infty &\equiv \tilde{V}_{++}(r \rightarrow \infty) = \frac{\Omega_-^2}{2\Delta_-} - \frac{\Omega_-^4}{8\Delta_-^3}, \\ \tilde{V}_{--}^\infty &\equiv \tilde{V}_{--}(r \rightarrow \infty) = \frac{\Omega_+^2}{2\Delta_+} - \frac{\Omega_+^4}{8\Delta_+^3}. \end{aligned} \quad (17)$$

The relative height of the potentials becomes

$$\begin{aligned} (\tilde{V}_{++} - \tilde{V}_{++}^\infty)/\tilde{V}_0 &= \frac{1 - \alpha_1^2 - \sigma\beta(r/R_1)^6}{\alpha_1^2 - [1 - \sigma(r/R_1)^6][1 - \beta\sigma(r/R_1)^6]}, \\ (\tilde{V}_{--} - \tilde{V}_{--}^\infty)/\tilde{V}_0 &= \left(\frac{\Omega_+}{\Omega_-}\right)^4 \frac{1}{\beta^3} \frac{1 - \alpha_1^2 - \sigma(r/R_1)^6}{\alpha_1^2 - [1 - \sigma(r/R_1)^6][1 - \beta\sigma(r/R_1)^6]}, \end{aligned} \quad (18)$$

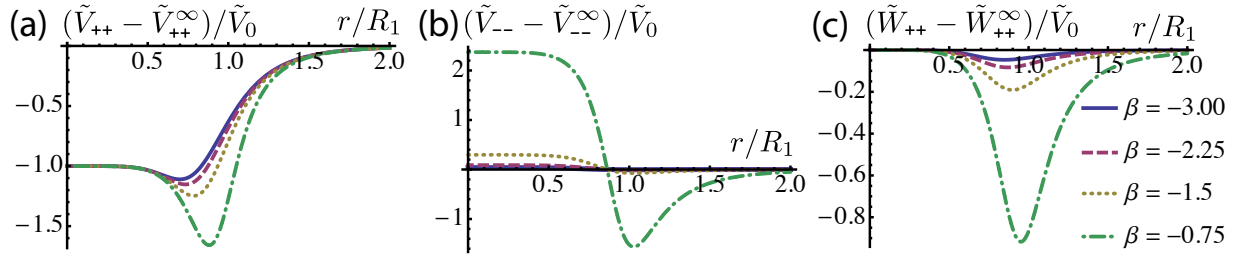


Figure 2. Plot of the relative height given by Eqs. (18) for $\alpha_1^2 = 1.41$ ($n = 60$) and $\sigma = -1$ for various laser detuning fractions β .

with $\alpha_1^2 = (w_{++}/c_{++})^2$ [shown in Fig. 2(b) in the main text], $\beta = \Delta_+/\Delta_-$, $\tilde{V}_0 = \Omega_-^4/(8\Delta_-^3)$, $\sigma = \text{sign}(c_{++})\text{sign}(\Delta_-)$ and $R_1^6 = |c_{++}|/(2|\Delta_-|)$. Due to the resolvent both potentials can be divergent for $W_{++}^2 - (V_{++} - 2\Delta_-)(V_{++} - 2\Delta_+) = 0$, when two Born-Oppenheimer surfaces undergo an avoided crossing. This happens at

$$R_{\text{div}}^6 = \frac{(\Delta_- + \Delta_+)c_{++} \pm \sqrt{(\Delta_- - \Delta_+)^2 c_{++}^2 + 4\Delta_- \Delta_+ w_{++}^2}}{4\Delta_- \Delta_+} = \frac{1 + \beta \pm \sqrt{(1 - \beta)^2 + 4\beta\alpha_1^2}}{2\beta} \sigma R_1^6. \quad (19)$$

In order to avoid such divergences and to obtain step-like potentials we require $\text{Im}(R_{\text{div}}) \neq 0$. For $\alpha_1^2 > 1$ this is for example the case when $\beta < 1 - 2\alpha_1^2 + 2\alpha_1\sqrt{\alpha_1^2 - 1}$. Figure 2 shows a typical example of Eq. (18) for $n = 60$ where $\alpha_1^2 = 1.41$ and $\sigma = -1$. In this case the potential has no singularity (avoided crossing) for $\beta < -0.30$.

For $\alpha_1 = 0$ one obtains the well known result of a single dressed Rydberg level, i.e. $(\tilde{V}_{++} - \tilde{V}_{++}^\infty)/\tilde{V}_0 = -1/[1 - \sigma(r/R_1)^6]$ [3].

C. Coupling element $\tilde{W}_{++}$

For the coupling matrix element $\tilde{W}_{++}$ adiabatic elimination up to fourth order in Ω/Δ yields

$$\tilde{W}_{++} = -e^{i\Delta\phi} \frac{\Omega_-^2 \Omega_+^2}{4\Delta_- \Delta_+} \frac{W_{++}}{W_{++}^2 - (V_{++} - 2\Delta_-)(V_{++} - 2\Delta_+)} \quad (20)$$

where $\Delta\phi = (\mathbf{k}_1 - \mathbf{k}_2)(\mathbf{r}_1 + \mathbf{r}_2)$ is the phase difference between the two lasers at the center of mass position. Note that this phase can be gauged away using a local gauge transformation – a rotation around the z-axis in the spin-basis. Asymptotically and at the origin ($r = 0$) the coupling matrix element vanishes

$$\tilde{W}_{++}(r \rightarrow \infty) = 0, \quad \text{and} \quad \tilde{W}_{++}(r \rightarrow 0) = 0. \quad (21)$$

In dimensionless units $\tilde{W}_{++}$ reads

$$\tilde{W}_{++}/\tilde{V}_0 = -\frac{1}{2\beta} \left(\frac{\Omega_+}{\Omega_-} \right)^2 \frac{\alpha_1 \sigma (r/R_1)^6}{\alpha_1^2 - [1 - \sigma(r/R_1)^6][1 - \beta\sigma(r/R_1)^6]}. \quad (22)$$

Again, this matrix element is regular for $\alpha_1^2 > 1$ and $\beta < 1 - 2\alpha_1^2 + 2\alpha_1\sqrt{\alpha_1^2 - 1}$.

Figure 2(c) shows a typical example of Eq. (22) for $n = 60$ where $\alpha_1^2 = 1.41$ and $\sigma = -1$. In this case the potential has no singularity (avoided crossing) for $\beta < -0.30$. The coupling matrix element has a maximum at

$$R_{1,\text{max}}^6 = \sqrt{(1 - \alpha_1^2)/\beta} R_1^6 \quad \text{with} \quad \tilde{W}_{++}(R_{1,\text{max}}) = -e^{i\Delta\phi} \frac{\tilde{V}_0}{2\beta} \left(\frac{\Omega_+}{\Omega_-} \right)^2 \frac{\alpha_1}{1 + 2\sqrt{\beta(1 - \alpha_1^2)} + \beta}. \quad (23)$$

D. Potential $\tilde{V}_{+-}$

For the potential $\tilde{V}_{+-}$ adiabatic elimination up to fourth order in Ω/Δ yields

$$\tilde{V}_{+-} = \frac{\Omega_-^2}{4\Delta_-} + \frac{\Omega_+^2}{4\Delta_+} - \frac{\Omega_-^4}{16\Delta_-^3} - \frac{\Omega_+^4}{16\Delta_+^3} - \frac{\Omega_-^2\Omega_+^2}{16\Delta_-^2\Delta_+} - \frac{\Omega_+^2\Omega_-^2}{16\Delta_+^2\Delta_-} - \frac{\Omega_+^4}{16\Delta_+^3} + \frac{(\Delta_- + \Delta_+)^2 \Omega_-^2 \Omega_+^2 (\Delta_- + \Delta_+ - V_{+-})}{16\Delta_-^2\Delta_+^2 ((\Delta_- + \Delta_+ - V_{+-})^2 - W_{+-}^2)} \quad (24)$$

where asymptotically we just recover the single particle light shifts (up to fourth order)

$$\tilde{V}_{+-}^\infty \equiv \tilde{V}_{+-}(r \rightarrow \infty) = \frac{\Omega_-^2}{4\Delta_-} - \frac{\Omega_-^4}{16\Delta_-^3} + \frac{\Omega_+^2}{4\Delta_+} - \frac{\Omega_+^4}{16\Delta_+^3}. \quad (25)$$

The relative height of the potential becomes

$$(\tilde{V}_{+-} - \tilde{V}_{+-}^\infty)/\tilde{V}_0 = \frac{(1+\beta)}{2\beta^2} \left(\frac{\Omega_+^2}{\Omega_-} \right)^2 \frac{\frac{1}{2}(1+\beta)\sigma'(r/R_2)^6 - 1 + \alpha_2^2}{[\frac{1}{2}(1+\beta)\sigma'(r/R_2)^6 - 1]^2 - \alpha_2^2} \quad (26)$$

with $\alpha_2^2 = (W_{+-}/c_{+-})^2$ [shown Fig.[2](b) of the main text], $\sigma' = \text{sign}(c_{+-})\text{sign}(\Delta_-)$ and $R_2^6 = |c_{+-}|/(2|\Delta_-|)$. Due to the resolvent the second term can be divergent for $(\Delta_- + \Delta_+ - V_{+-})^2 - W_{+-}^2 = 0$, when two potential surfaces undergo an avoided crossing. This happens at

$$R_{\text{div}}^6 = \frac{c_{+-} \pm w_{+-}}{\Delta_- + \Delta_+} = \frac{1 \pm \alpha_2}{\frac{1}{2}(1+\beta)} \sigma' R_2. \quad (27)$$

In order to avoid such divergences and to obtain step-like potentials we require $R_{\text{div}} \in \mathbb{C}$. This can only be fulfilled for $-1 < \alpha_2 < 1$ and $\beta < -1$. Figure 3 shows a typical example of Eq. (26) for $n = 60$ where $\alpha_2^2 = 0.46$ and $\sigma' = 1$. In this case the potential has no singularity (avoided crossing) for $\beta < -1$. We note that for $\beta = -1$ the potential vanishes.

E. Coupling element $\tilde{W}_{+-}$

For the coupling matrix element $\tilde{W}_{+-}$ adiabatic elimination up to fourth order in Ω/Δ yields

$$\tilde{W}_{+-} = e^{i\Delta\phi_{12}} \frac{\Omega_-^2\Omega_+^2}{16\Delta_-^2\Delta_+^2} \frac{(\Delta_- + \Delta_+)^2 W_{+-}}{(\Delta_- + \Delta_+ - V_{+-})^2 - W_{+-}^2} \quad (28)$$

where $\Delta\phi_{12} = (\mathbf{k}_1 - \mathbf{k}_2)(\mathbf{r}_1 - \mathbf{r}_2)$ is the phase difference between the two lasers and relative position. Asymptotically and at the origin ($r = 0$) the coupling matrix element vanishes

$$\tilde{W}_{+-}(r \rightarrow \infty) = 0, \quad \text{and} \quad \tilde{W}_{+-}(r \rightarrow 0) = 0. \quad (29)$$

In dimensionless units $\tilde{W}_{+-}$ reads

$$\tilde{W}_{+-}/\tilde{V}_0 = \left(\frac{\Omega_+}{\Omega_-} \right)^2 \frac{1}{2\beta^2} \frac{\alpha_2 \frac{1}{2}(1+\beta)^2 \sigma'(r/R_2)^6}{[\frac{1}{2}(1+\beta)\sigma'(r/R_2)^6 - 1]^2 - \alpha_2^2}. \quad (30)$$

Again, this matrix element is regular for $-1 < \alpha_2 < 1$ and $\beta < -1$. Figure 3 (right panel) shows a typical example of Eq. (30) for $n = 60$ where $\alpha_1^2 = 1.41$ and $\sigma = -1$. We note that for $\beta = -1$ the coupling matrix element vanishes. The coupling matrix element has a maximum at

$$R_{2,\text{max}}^6 = -\frac{2\sqrt{1-\alpha_2^2}}{1+\beta} R_2^6 \quad \text{with} \quad \tilde{W}_{+-}(R_{2,\text{max}}) = -\frac{\tilde{V}_0}{4\beta^2} \left(\frac{\Omega_+}{\Omega_-} \right)^2 \left(\sqrt{1-\alpha_2^2} - 1 \right) \frac{\beta+1}{\alpha_2}. \quad (31)$$

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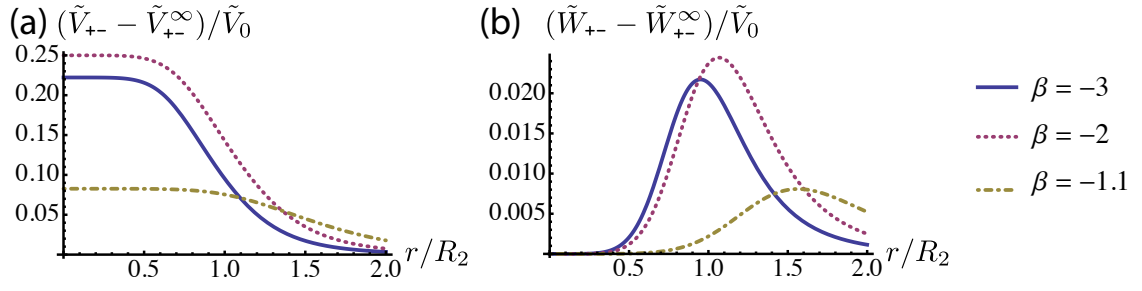


Figure 3. We plot the relative height of $\tilde{V}_{+-}$ given by Eq. (26) (left panel) and of $\tilde{W}_{+-}$ given by Eq. (30) (right panel) for $\alpha_2^2 = 0.46$ ($n = 60$), $\Omega_- = \Omega_+$ and $\sigma' = 1$ for various laser detuning fractions β .

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